

1. M/J/22/22/Q1

Fig. 1.1 shows a waterskier pulled horizontally across the water surface.



Fig. 1.1

The mass of the waterskier is 60 kg. The gravitational field strength g is 10 N/kg.

(a) Calculate the weight of the waterskier.

weight = [1]

(b) Mass and weight have different units.

State **two** other differences between mass and weight.

.....
.....
.....
..... [2]

2. O/N/21/22/Q1

Fig. 1.1 shows a jug of liquid and an empty measuring cylinder that is on an electronic balance.

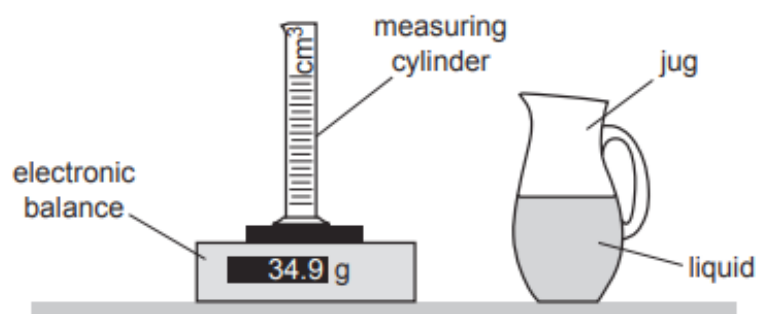


Fig. 1.1

- (a) The electronic balance uses the weight of the measuring cylinder to determine its mass.

Weight and mass have different units.

State **two** other ways in which weight differs from mass.

1.
2.

[2]

- (b) Describe how to determine the density of the liquid using the apparatus shown in Fig. 1.1.

.....
.....
.....
.....
.....
.....
.....

[4]

- (c) The density of the liquid is 780 kg/m^3 .

Determine the volume of 0.65 kg of the liquid.

volume = [2]

Total: 8

3. O/N/19/21/Q5

An electronic balance is calibrated to display the weight of an object that is placed on its top plate.

Fig. 5.1 shows the balance with an empty measuring cylinder on the top plate.

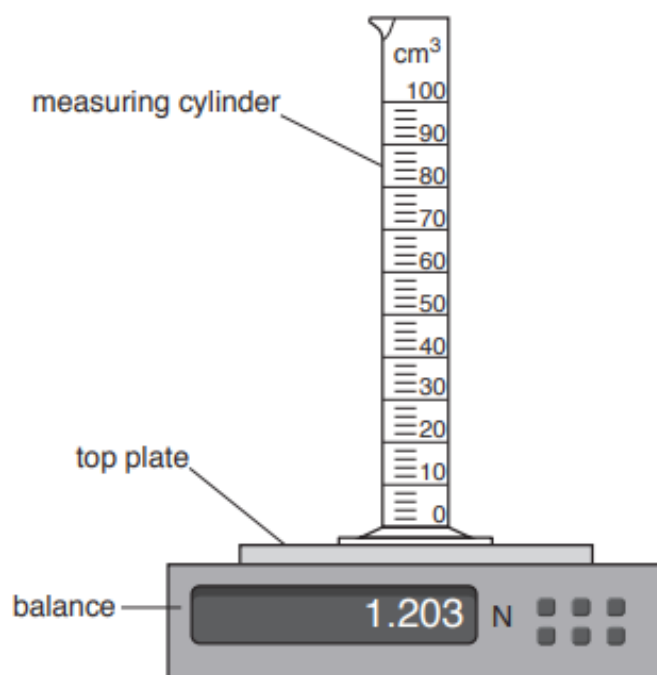


Fig. 5.1

A student pours a small quantity of oil into the measuring cylinder and records the volume and the new reading on the balance.

Fig. 5.2 shows how the reading on the balance varies with the volume of oil in the measuring cylinder.

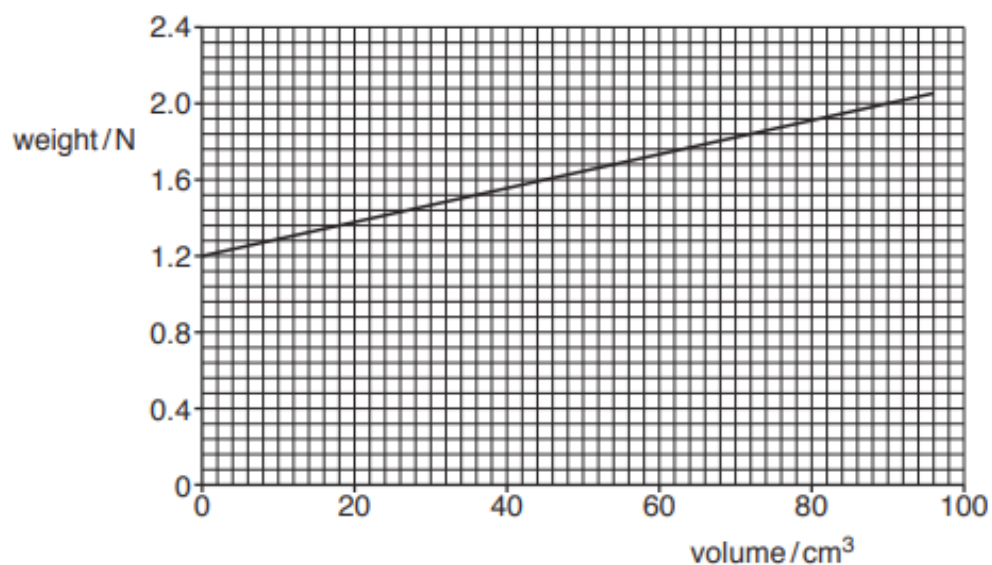


Fig. 5.2

The gravitational field strength g is equal to 10 N/kg .

Using Fig. 5.2, determine the density of the oil.

density = [4]

4. M/J/19/22/Q1

A student records the mass and volume of two irregular objects, one made of iron and the other of copper. The student uses the equipment shown in Fig. 1.1.

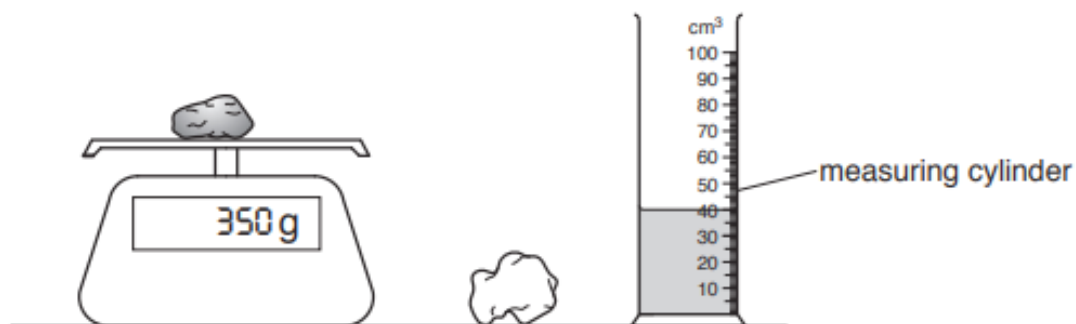


Fig. 1.1

Table 1.1 shows the results obtained.

Table 1.1

	mass / g	volume / cm ³
iron	400	51
copper	350	39

(a) Describe how to determine the volume of an irregular object with the measuring cylinder.

.....

.....

.....

.....

..... [2]

(b) Calculate the density of iron.

density = [2]

- (c) A third object, which is also made of copper, has the same volume as the iron object.

Calculate the mass of this copper object.

mass = [2]

- (d) State and explain what happens to the density of the iron object when it is heated.

.....
.....
..... [1]

[Total: 7]

5. M/J/18/22/Q3

Equal volumes of metal, water and air are heated from 20 °C to 80 °C.

- (a) State whether the metal, the water or the air expands the most.

.....[1]

- (b) Fig. 3.1 is a diagram of the metal at 20 °C. The metal is a large, square sheet with a square hole at its centre. The sheet lies flat on a table.

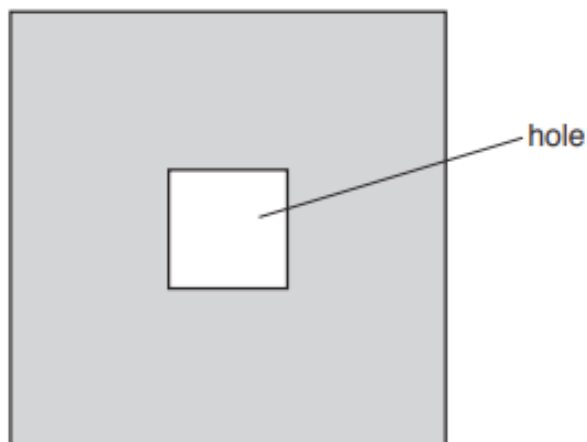


Fig. 3.1

On Fig. 3.1, draw the outline of the sheet and of the hole after the metal is heated. Make sure that the expansion is noticeable. [2]

- (c) The metal has a mass of 5.0 kg and a density of $7.5 \times 10^3 \text{ kg/m}^3$.

- (i) Define *density*.

.....
.....[1]

- (ii) Calculate the volume of the metal.

Give your answer to a suitable number of significant figures.

volume = [2]

6. M/J/11/22/Q1

A student wishes to find the density of a stone. He uses a measuring cylinder and a spring balance with a scale marked in newtons. The measuring cylinder, spring balance and stone are shown in Fig. 1.1.

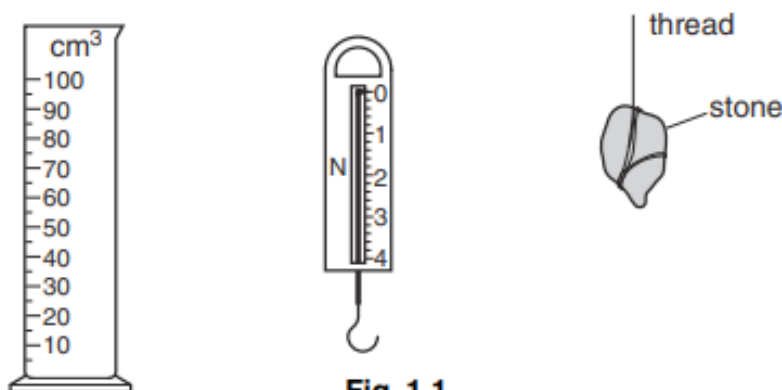


Fig. 1.1

The student knows that the gravitational field strength is 10 N/kg.

- (a) Describe how the student uses the spring balance to find the mass of the stone.

.....

[2]

- (b) Describe how the student uses the measuring cylinder to find the volume of the stone.

.....

[2]

- (c) The mass of the stone is 150 g and its volume is 70 cm³.
 Calculate the density of the stone.

density of stone =[1]

- (d) The stone is taken to another place, where the gravitational field strength is less than 10 N/kg. State how this affects the mass and the weight of the stone.

mass

weight

[1]

7. M/J/16/21/Q3

Fig. 3.1 shows a thin sheet of plastic. A student tries to measure the thickness of the sheet with a ruler, but the sheet is too thin to measure accurately.

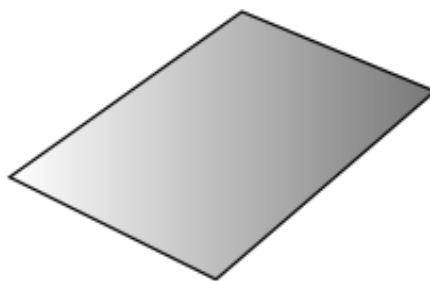


Fig. 3.1 (not to scale)

The student measures the mass of the sheet and obtains the value 0.12 g.

(a) (i) State what is meant by *mass*.

.....
.....[1]

(ii) The student is told that the density of the plastic is 0.91 g/cm^3 .

Calculate the volume of the plastic sheet.

volume =[2]

(iii) The student measures the length and width of the sheet. The readings obtained are:

length of sheet = 3.0 cm

width of sheet = 2.0 cm

Calculate the thickness of the sheet.

thickness = [2]

(b) State a measuring instrument that can be used to measure the thickness of the sheet accurately.

.....[1]

8. O/N/14/22/Q3

Fig. 3.1 shows a helium cylinder being used at a funfair to inflate balloons that float in the air.

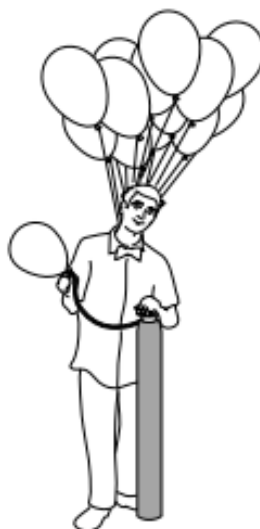


Fig. 3.1

Initially, all the helium is stored in the cylinder at a pressure of $2.0 \times 10^5 \text{ Pa}$.

When no more balloons can be filled, all the helium occupies a total volume of 1.8 m^3 and is at a pressure of $1.0 \times 10^5 \text{ Pa}$. The temperature of the helium has not changed.

- (a) Calculate, stating the formula used, the initial volume of the helium in the cylinder.

volume =[3]

- (b) The balloon seller takes the cylinder back to the helium supplier and it is refilled. The mass of helium in the cylinder is 0.30 kg .

- (i) Calculate the density of helium in the cylinder.

density =[2]

- (ii) The mass of the empty cylinder is 16 kg .

Suggest why weighing the cylinder is not an accurate method of determining the amount of helium supplied.

.....
[1]

9. O/N/14/21/Q2

A student places a metre rule on the edge of a triangular prism. The prism is used as a pivot and the rule balances about the 50 cm mark.

The student then places a block of wood at the 10 cm mark on the rule and an empty measuring cylinder at the 80 cm mark. The rule is still balanced. Fig. 2.1 shows the arrangement.

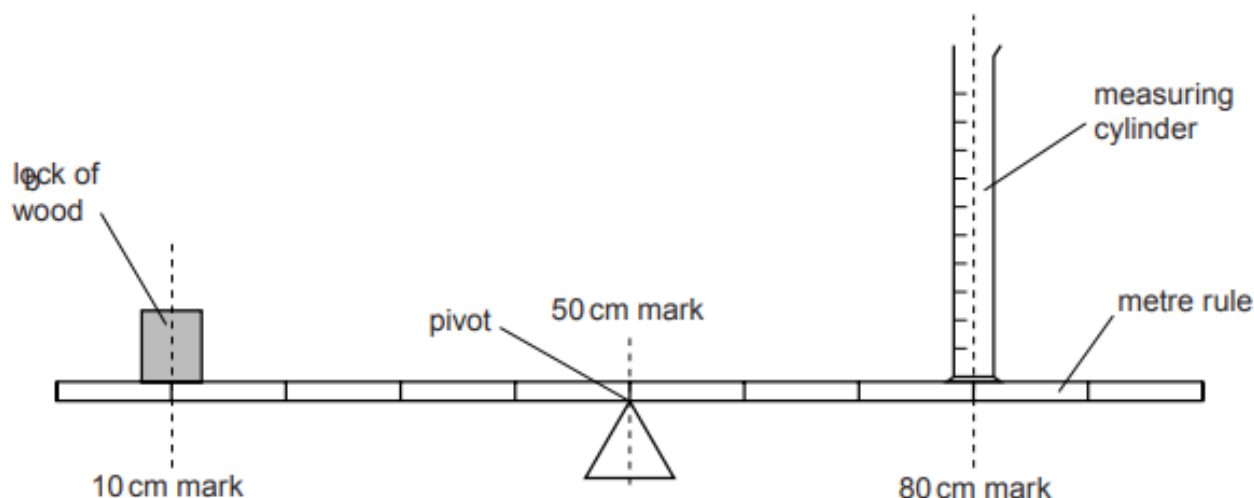


Fig. 2.1 (not to scale)

The student now places a weight of 0.39 N on top of the block of wood. She then starts to pour oil into the measuring cylinder. The rule balances again when there is 60 cm³ of oil in the measuring cylinder.

The gravitational field strength g is 10 N/kg.

(a) Calculate

(i) the weight of the oil in the cylinder,

weight =[2]

(ii) the mass of the oil in the cylinder.

mass =[1]

(b) Calculate the density of the oil.

density =[2]

10. O/N/13/22/Q2

Fig. 2.1 shows a boy moving a water container in a wheelbarrow.

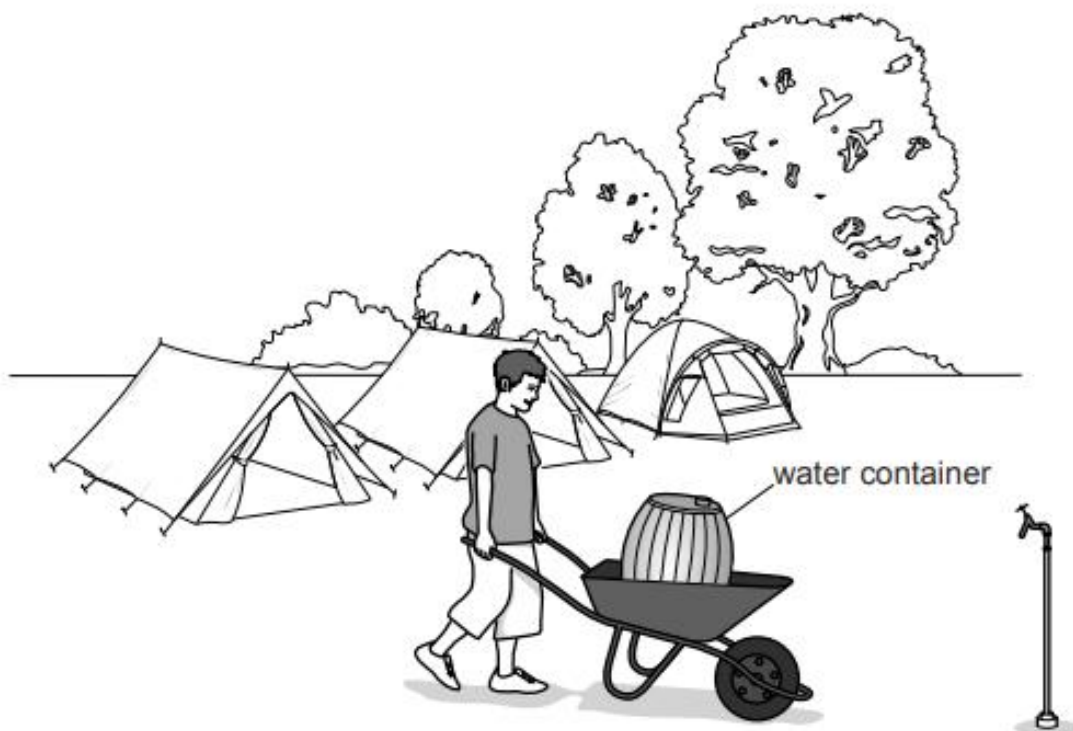


Fig. 2.1

The container has a volume of 0.15 m^3 and is filled with water of density 1000 kg/m^3 .

(a) Calculate the mass of water in the container when it is full.

mass =[2]

(b) It is harder to stop the wheelbarrow when the container is full than when it is empty.

Explain this.

.....
.....
.....[2]

11. M/J/11/22/Q1

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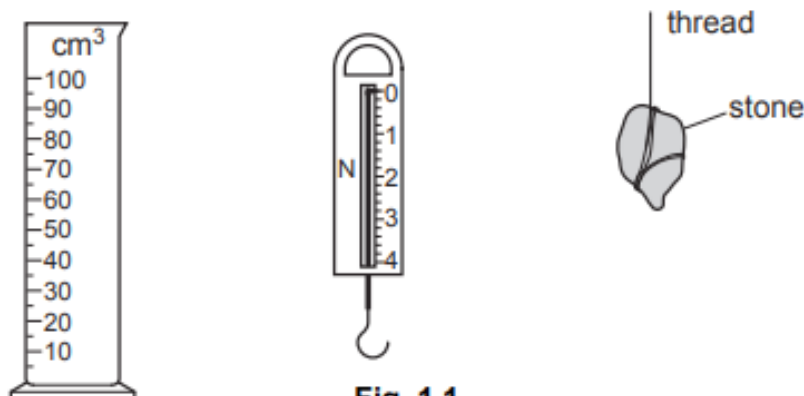


Fig. 1.1

The student knows that the gravitational field strength is 10 N/kg .

(a) Describe how the student uses the spring balance to find the mass of the stone.

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.....[2]

(b) Describe how the student uses the measuring cylinder to find the volume of the stone.

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.....

.....[2]

(c) The mass of the stone is 150 g and its volume is 70 cm^3 .
Calculate the density of the stone.

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mass

weight

[1]